SECTION - D

Explain in detail

1. Write the specification of an algorithm hypotenuse whose inputs are the lengths of the two shorter sides of a right angled triangle, and the output is the length of the third side.

2. Suppose you want to solve the quadratic equation ax2 + bx + c = 0 by an algorithm.

   quadratic_solve (a, b, c)

   -- **inputs : ?**

   -- **outputs: ?**

   You intend to use the formula and you are prepared to handle only real number roots. Write a suitable specification.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

3. Exchange the contents: Given two glasses marked A and B. Glass A is full of apple drink and glass B is full of grape drink. For exchanging the contents of glasses A and B, represent the state by suitable variables, and write the specification of the algorithm.

Books

1. Roland Backhouse, Algorithmic Problem Solving, John Wiley & Sons Ltd, 2011.

2. Krysia Broda, Susan Eisenbach, Hessam Khoshnevisan, Steve Vickers, Reasoned Programming, Prentice Hall, 1994

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